Econ 414: Discussion Section 6

GSI: Brian Daza
bdaza@umich.edu

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University of Michigan

Today's Discussion

Review the Schumpeterian model:

- Why do firms earn temporary monopoly profits?
- How does limit pricing work?
- Why does innovation destroy existing firms?
- Why does creative destruction affect valuation?
- How does free entry determine s_R?

Technical checklist:

- Micro-structure and price setting.
- Growth and valuation.

Question 1: Limit Pricing and Profits

Leader firm sets:

$$p = (1 + \gamma)r$$

Profits per firm:

$$\pi = \frac{\gamma}{1 + \gamma} \frac{\alpha}{J} Y$$

(a) Why can the leader not charge an arbitrarily high price?

(b) What role does the fringe firm play?

(c) How is the markup determined in this model?

Question 2: Growth of Profits

Profits:

$$\pi_t = \frac{\gamma}{1+\gamma}\frac{\alpha}{J}Y_t$$

(a) Show that $g_\pi = g_Y$.

(b) Using $g_Y = g_L + g_A$, what determines profit growth?

(c) How does this differ from the Romer model?

Question 3: Creative Destruction and Valuation

Value of innovation:

$$V = \frac{\pi}{r - g_\pi + E[dN]}$$

(a) Why does $E[dN]$ appear in the denominator?

(b) What is the "effective discount rate"?

(c) Why is monopoly temporary in this model?

Innovation increases profits — but also shortens their duration.

Mechanism Mapping: Increase in γ

Trace the effects of a larger step size γ:

1. What happens to markup?
2. What happens to profits π?
3. What happens to V?
4. What happens to s_R?
5. What happens to growth $g_A = \gamma E[dN]$?

Mechanism Mapping: Increase in $\mathbb{E}[dN]$

Recall:

$$V = \frac{\pi}{r - g_\pi + \mathbb{E}[dN]}$$

What happens if $\mathbb{E}[dN]$ increases?

1. How does this affect valuation?
2. Why does it reduce monopoly duration?
3. Why does it also increase innovation incentives?
4. Which effect dominates?

Dual role of innovation: creation vs destruction.

Question 4: Free Entry and R&D

Free entry:

$$\frac{\pi}{r - g_\pi + E[dN]} = F$$

Which implies:

$$\frac{s_R}{1 - s_R} = \frac{\alpha \frac{\gamma}{1+\gamma}}{1 - \alpha} \frac{E[dN]}{r - g_L + E[dN](1-\gamma)}$$

Without algebra, discuss:

(a) What happens if r increases?

(b) What happens if g_L increases?

(c) How does this compare to Romer?

Three Core Takeaways

1. Monopoly power is temporary.
2. Innovation both creates and destroys profits.
3. Growth comes from quality upgrading, not variety expansion.

Creative destruction is the engine of growth.

Self-Assessment

Without notes, could you:

1. Explain why monopoly profits are temporary?
2. Derive $g_A = \gamma E[dN]$?
3. Explain why $E[dN]$ both encourages and discourages R&D?
4. Trace the effect of a higher γ on growth?

Romer vs. Schumpeter

This table summarizes how the models covered in the last two weeks compare:

	Romer	Schumpeter
Number of goods	Expands	Fixed
Innovation	New variety	Quality jump
Monopoly	Permanent	Temporary
Profit growth	g_L	$g_L + g_A$
Destruction	No	Yes

Technical Checklist I: Micro Structure

To fully manipulate the Schumpeterian model, you should be comfortable with:

- Deriving intermediate firm revenues:

$$p_j x_j = \frac{\alpha}{J} Y$$

- Understanding limit pricing:

$$p = (1+\gamma) r$$

  and why markup is determined by step size γ.

- Computing profits:

$$\pi = \frac{\gamma}{1+\gamma} \frac{\alpha}{J} Y$$

- Explaining why monopoly power is temporary.
- Understanding the quality ladder:

$$a(N_j) = (1+\gamma)^{N_j}$$

Technical Checklist II: Growth and Valuation

You should be comfortable with:

- Computing aggregate technology growth:

$$g_A = \gamma E[dN]$$

- Showing that profit growth satisfies:

$$g_\pi = g_Y = g_L + g_A$$

- Continuous-time valuation with creative destruction:

$$V = \int_0^\infty \pi_0 e^{(g_\pi - r - E[dN])t} dt = \frac{\pi_0}{r - g_\pi + E[dN]}$$

- Interpreting $E[dN]$ as an additional discount term.
- Using free entry to determine s_R.

Game-Theoretic Structure of Limit Pricing

Players:

- Leader (quality $a(N_j)$)
- Fringe (quality $a(N_j - 1)$)

Timing:

1. Leader chooses price p_L.
2. Fringe decides whether to enter.

Fringe can enter profitably if:

$$p_L \geqslant (1+\gamma)r$$

Leader chooses:

$$p_L^* = (1+\gamma)r$$

Highest price that deters entry.

Limit Pricing as a Game: Payoff Matrix (Stylized)

Interpretation:

- Leader chooses a pricing *regime*:
  - **Limit price** (deter entry): $p_L = (1+\gamma)r$
  - **High price** (try to extract more): $p_L > (1+\gamma)r$
- Fringe chooses: **Enter** or **Stay Out**

	Fringe: Stay Out	**Fringe: Enter**
Leader: Limit price	$(\pi^L, 0)$	$(\pi^L, \leqslant 0)$
Leader: High price	$(\pi^H, 0)$	(π^E, π^F)

- $\pi^H > \pi^L$: if no entry, leader prefers charging high price.
- But if leader charges high price, fringe can profitably enter: $\pi^F > 0$.
- Under limit pricing, fringe cannot profitably enter: payoff $\leqslant 0$.

Main message: limit pricing trades off lower price today for deterring entry.

Limit Pricing as a Sequential Game (Extensive Form)

Players: Leader (quality $a(N_j)$) and Fringe (quality $a(N_j - 1)$).
Marginal cost for both: $MC = r$. Fringe can profitably undercut only if $p_L > (1+\gamma)r$.

